

ANIRUDH BHATTACHARYA

Application Engineer, Physical Design | Fusion Compiler / ICC2 | PD CAD

Hyderabad | Portfolio: anirudhbhattaacharya.github.io/portfolio

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EXPERIENCE

Synopsys Inc.

Physical Design Application Engineer

Apr 2025 to Present

Hyderabad, Telangana

- Debugged live Fusion Compiler and ICC2 PnR databases for product teams across floorplan, placement, legalization, CTS / MSCTS, routing, timing closure, MCMM STA, multi-voltage UPF, IR / EM, congestion and Calibre physical verification / sign-off on TSMC N5A, 40ULP, 180G and Intel 18A / IFS18A.
- Isolated a live Qualcomm PnR fail across placement, CTS and route and shipped GSAE testcase `gsaetc_fc_command_trace` from `sh_command_log_file` and `log_trace`. Synopsys Excellence Award, March 2026. Qualcomm GSAE testcase reused across PD and AE teams.
- Closed 100+ Fusion Compiler and ICC2 cases (~6 per month) for Qualcomm, Samsung, Microsoft, Sony, ST, Marvell, Toshiba, Arm, NXP and MediaTek.
- Handled live escalations from Qualcomm and Samsung product PD teams on production PnR and physical-verification fails.
- Resolved production breaks in `compile_fusion`, MSCTS, useful skew / CCD, N5A congestion and legalization, via-ladder routing, UPF split-fanout and Calibre LVS sign-off.
- Wrote production Tcl reused on repeat customer fails: ECO displacement CSV, cells-outside-core with ray-cast boundary, legalizer CMD-005 isolation, and Qualcomm command-trace extraction.
- Worked with customer PD teams and internal AE peers to reproduce N5A DRC / LVS and congestion fails and publish the workaround flow.

Synopsys Inc.

Physical Design Intern

Apr 2024 to Mar 2025

Hyderabad / Bhubaneswar

- Completed Fusion Compiler RTL-to-GDS training covering floorplan, PDN, CTS, PnR, timing closure, STA and Calibre physical verification.
- Ran DSO.ai with Fusion Compiler on SLM PVT Monitor IP at Intel 18A / IFS18A (process detector, voltage monitor, temperature sensor) for timing, power and block-level QoR.

VLSI Expert Private Limited

VLSI Trainee, Physical Design

May 2022 to Feb 2023

Noida, Uttar Pradesh

- Took RISC-V picorv32a from Verilog through floorplan, PDN, CTS, PnR and MCMM STA in ICC2 and PrimeTime, including SDC and max / min parasitic corners.

SELECTED WORK

Work	Node / tool	What I shipped	Result
Qualcomm PnR GSAE testcase	FC, Tcl	Command-trace of a live PnR fail	Reused; Award Mar 2026
Legalizer CMD-005 testcase	FC / ICC2	Isolated prelegal callback option	Two GSAE-style cases
Customer FC / ICC2 queue	N5A, 40ULP, 18A	CTS, legalize, route, UPF, PV	100+ cases
SLM PVT Monitor IP	Intel 18A / IFS18A	DSO.ai + Fusion Compiler QoR	Block-level PPA

SOLVNETPLUS: 30+ PUBLISHED ARTICLES

- Published 30+ SolvNetPlus articles from live FC / ICC2 cases, including CTS balancing and clock skew group pins, CCD prepone / postpone limits, IP-edge congestion, DRC convergence in `compile_fusion`, antenna and diode insert, crosstalk on `reg2out`, and unconstrained ICG paths DC vs PrimeTime.

TECHNICAL SKILLS

Languages: Tcl, Python, Perl, Make, Verilog, Linux

CAD / EDA: Fusion Compiler, ICC2, PrimeTime, Design Compiler, DSO.ai, Calibre

Physical Design: Floorplan, Placement, Legalize, CTS / MSCTS, PnR, PDN, timing closure, STA / MCMM, multi-voltage UPF, IR / EM, congestion, physical verification, sign-off, SDC, useful skew / CCD

Nodes: TSMC N5A, 40ULP, 180G; Intel 18A / IFS18A

EDUCATION

Jaypee University of Engineering and Technology

B.Tech, Electronics and Communications Engineering

2019 to 2023

Guna, Madhya Pradesh